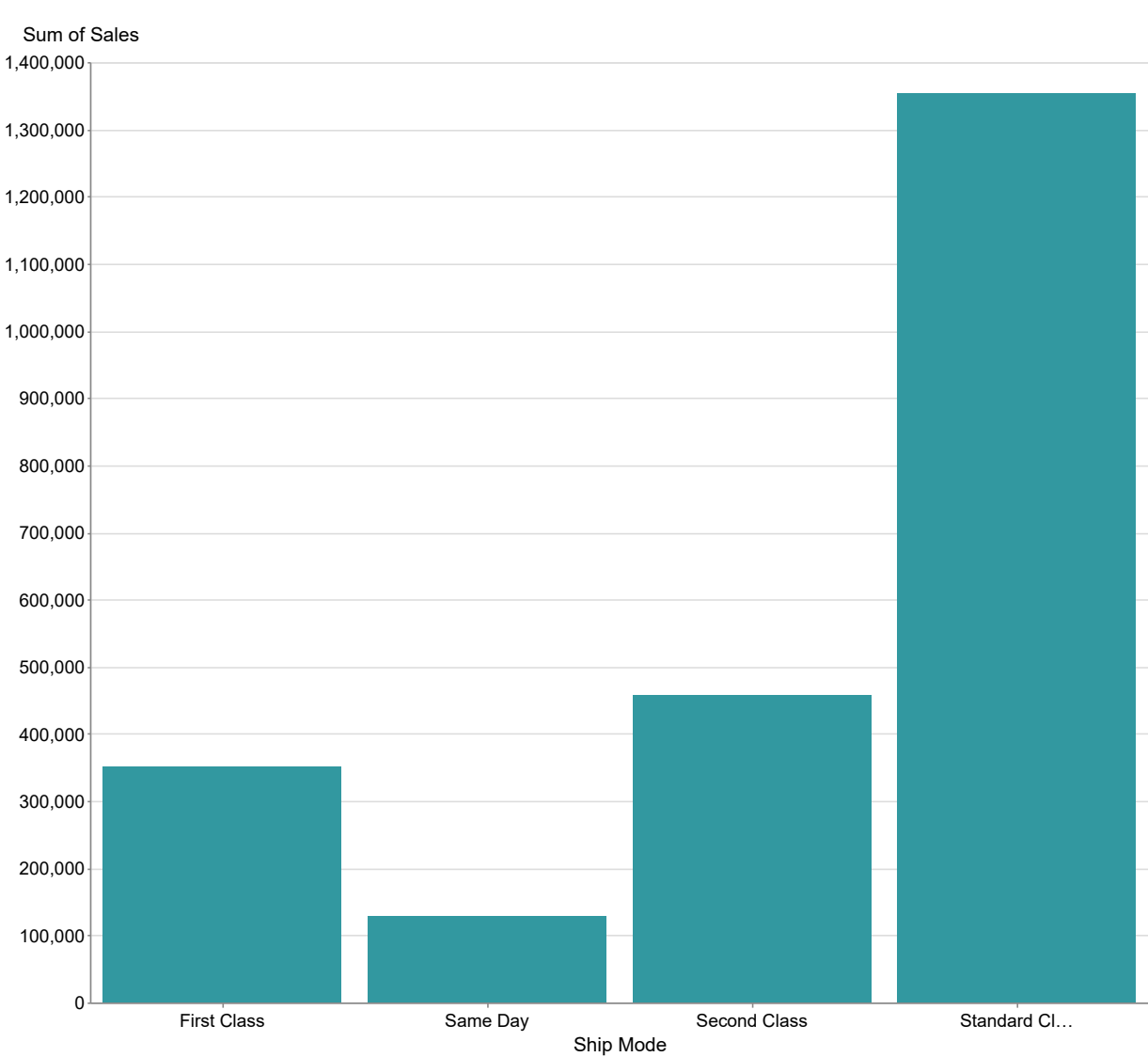
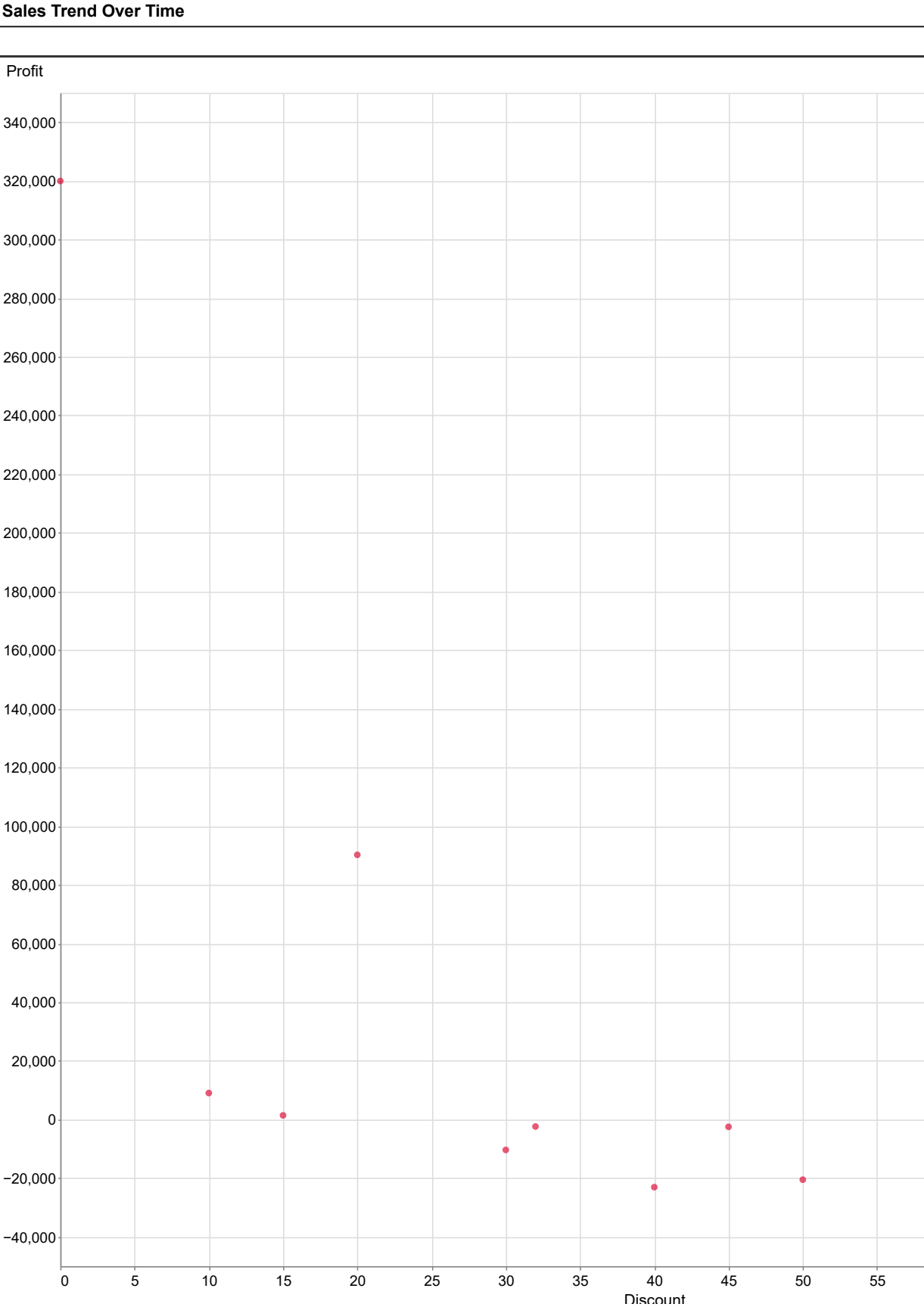
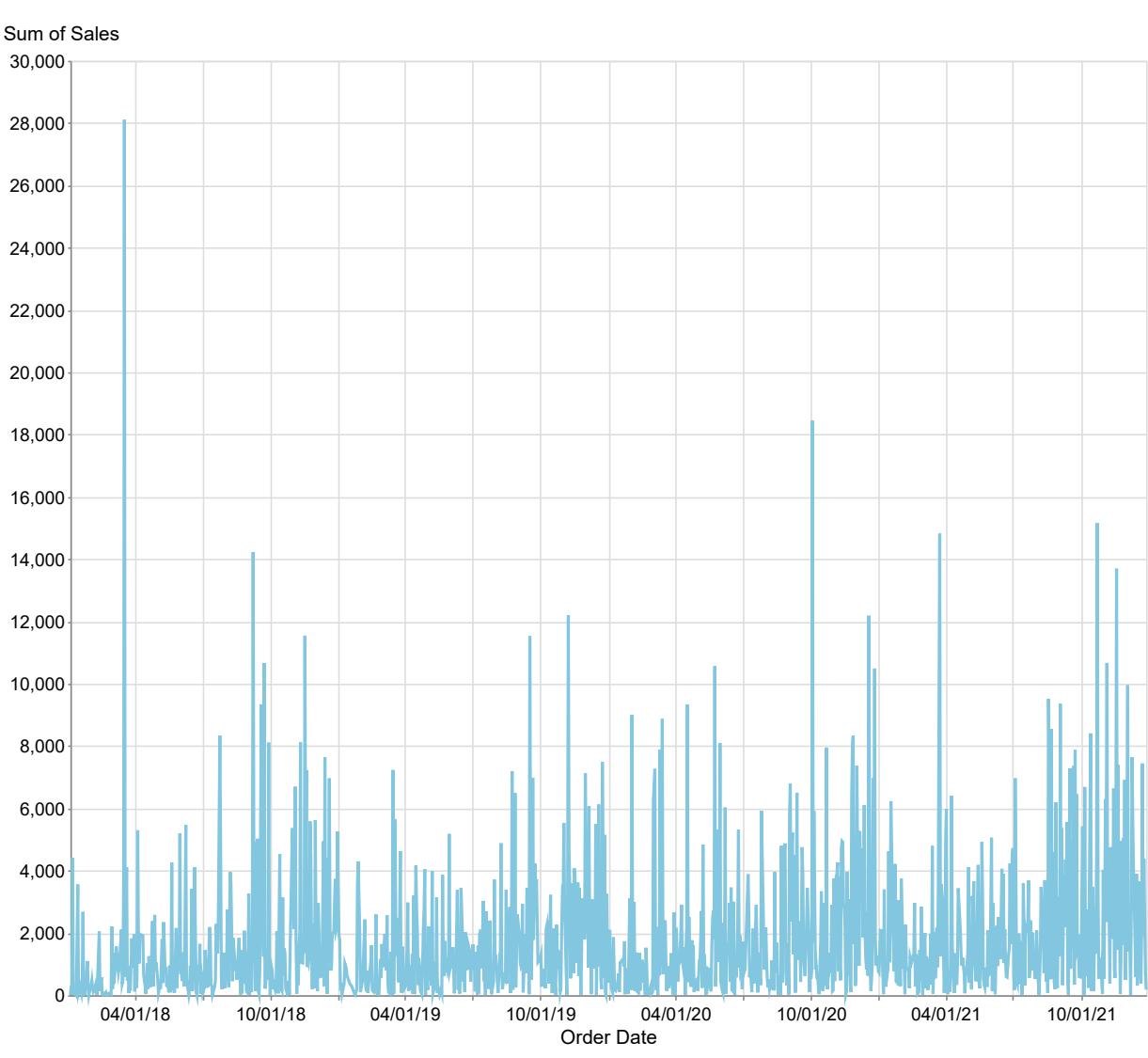


Which shipping modes are used most?

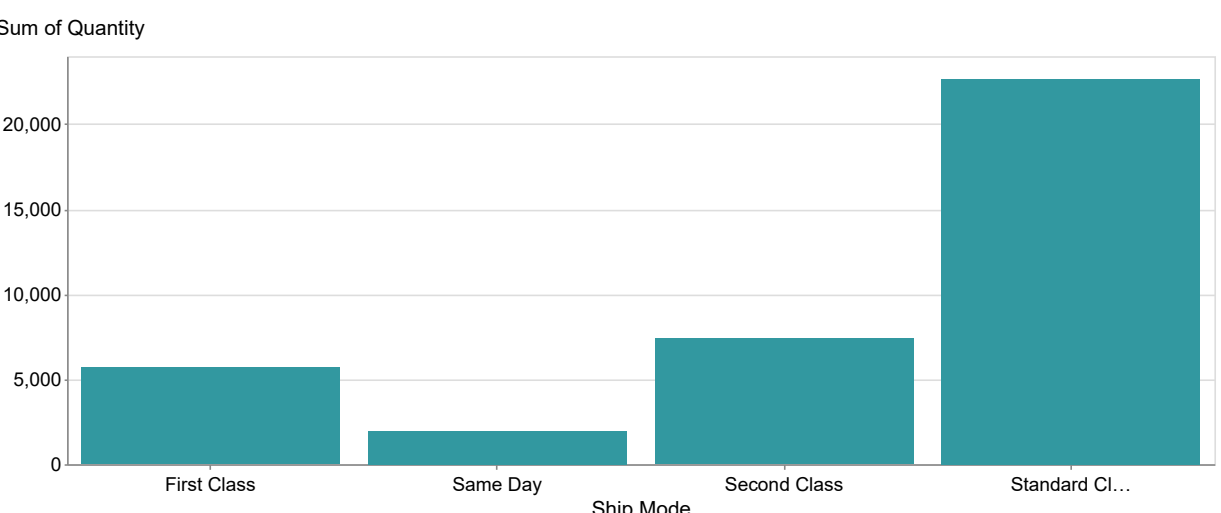
The sales data reveals that the Standard Class mode generated the highest sales at approximately \$1,353,342, followed by Second Class at about \$455,052, First Class at around \$351,082, and Same Day at \$129,343. The significant disparity between Standard Class and the other modes indicates a potential preference for this shipping option, while Same Day, being notably lower sales, suggesting it may not be as popular or necessary. Further analysis is needed, particularly with Same Day, which is significantly lower than the other categories, indicating a potential area for improvement or investigation. Future developments could focus on enhancing the appeal of Same Day shipping to capture a larger market share.



The dataset contains 1,235 orders of sales data spanning from January 1, 2016, to September 17, 2021, with sales figures ranging from approximately \$40.54 to \$4,848.87. Notably, the highest sales figure of \$4,848.87 occurred on September 8, 2021, while the lowest was \$40.54 on January 10, 2016, indicating significant growth over time. The data shows a general upward trend in sales, particularly in 2021, with a peak of \$4,848.87 on September 8, 2021, and a notable dip in sales in early 2020, likely influenced by external factors such as the COVID-19 pandemic. Future developments may continue to reflect this upward trajectory, but monitoring for potential anomalies will be essential to understand market dynamics.



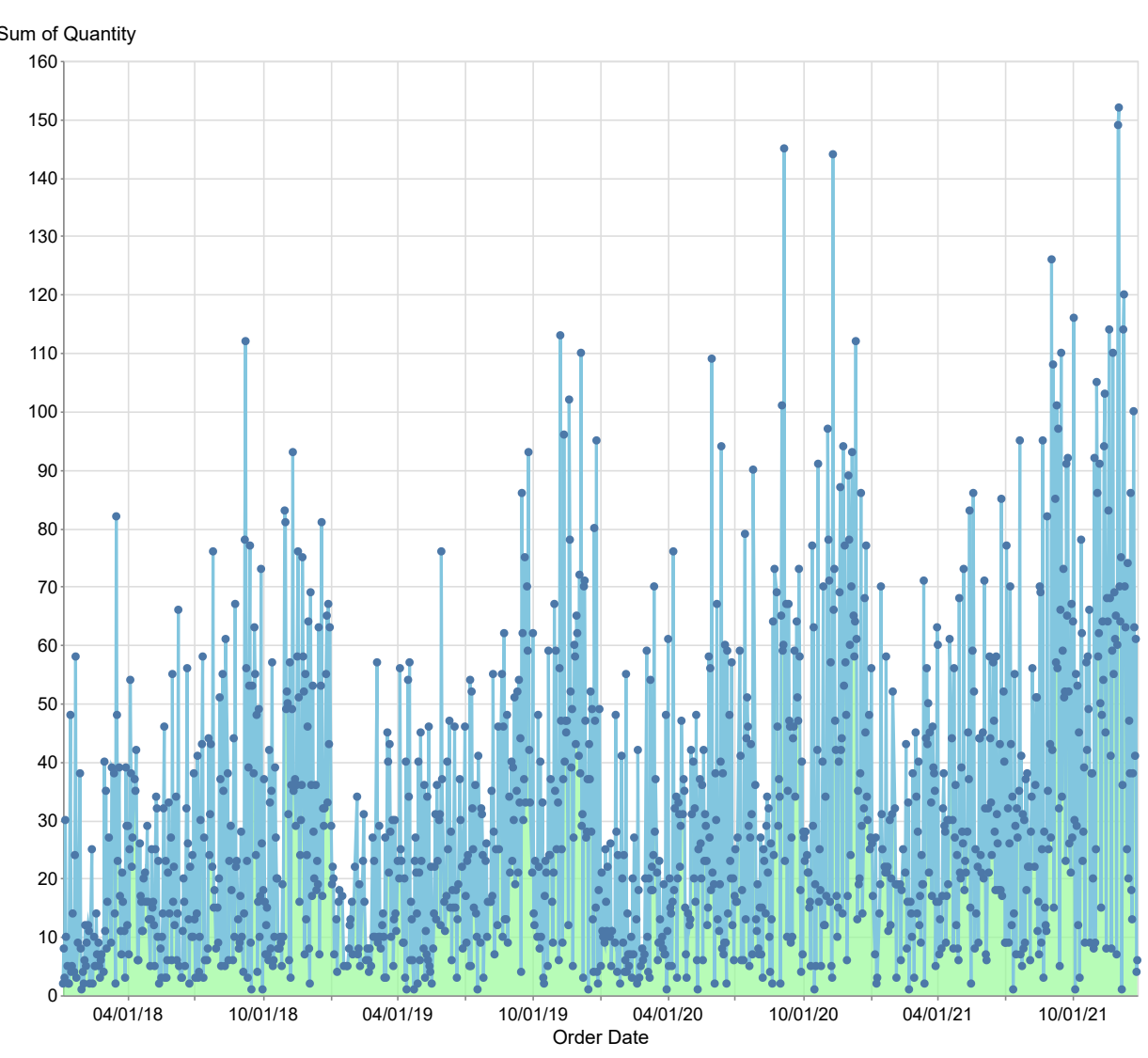
The data shows that Standard Class was the highest quantity at 22,651, followed by Second Class with 9,403, First Class with 8,649, and Same Day with 1,760. The significant disparity between Standard Class and Same Day indicates a potential preference for more economical shipping options, while Same Day appears to be an outlier with a notably lower quantity. The spread of quantities suggests a concentration of orders in the Standard and Second Class categories, which may indicate a trend towards cost-effective shipping solutions. Future developments could focus on enhancing Same Day services to capture a larger market share, addressing the anomaly in its low volume.



The data reveals that the Standard Class shipping mode generated the highest profit at \$455,052.94, significantly outperforming the other modes. While Second Class followed at \$351,082.25, First Class at \$455,052.94, and Same Day at \$129,343.43. The spread of profits indicates a notable disparity, particularly between Same Day and Standard Class, suggesting potential inefficiencies or market preferences. Anomalies are evident, as Same Day's profit is disproportionately low compared to its operational costs, which may warrant further investigation. Future developments could focus on optimizing Same Day services to ensure profitability and exploring the factors contributing to the dominance of Standard Class.



The dataset contains 1,235 orders of order quantities over various dates, with a notable peak in quantity on September 8, 2021, reaching 85 units, while the lowest recorded quantity is 3 units on January 10, 2016. The average quantity ordered is approximately 29.5 units, with a standard deviation of about 18.4, indicating a moderate spread in order quantities. Significant anomalies include the high order on September 8, 2021, and a low of 4 units on January 10, 2016, suggesting potential seasonal trends or external factors influencing order behavior.



The data shows a clear inverse relationship between discount percentages and profit. As profits increase significantly as discounts increase beyond 10%. Notably, the highest profit occurs at a 0% discount (\$4,848.87), while the lowest profit is at a 10% discount (\$40.54), indicating a potential drop of 3,797.29. The spread of profits becomes negative after a 10% discount, with significant anomalies at 30% (-10,366.04) and 40% (-23,057.05), where profits drop sharply. A potential mathematical model to describe this relationship could be a quadratic regression, suggesting a formula of the form $P = aD^2 + bD + c$, where P is profit, D is discount, and a , b , c are constants determined through regression analysis.